

Table of Contents

Scope of Work — Play School Website & Mobile Application

Project Name: Play School Digital Learning & Engagement Ecosystem **Document Type:** Master Scope of Work (SoW) **Version:** 1.0 **Date:** 2026-05-20 **Prepared For:** Play School Management **Prepared By:** Infiniti Tech Partners

Table of Contents

- 1. Project Overview
 - 2. Project Vision & Goals
 - 3. Master Feature List
 - 4. Technology Stack
 - 5. Software Architecture
 - 6. Data Flow
 - 7. API Reference
 - 8. Route Map (Frontend + Backend)
 - 9. Folder Structure
 - 10. User Roles & Permissions
 - 11. Database Design Overview
 - 12. Security & Compliance
 - 13. Performance & Scalability
 - 14. Accessibility & Localization
 - 15. Third-Party Integrations
 - 16. Deliverables
 - 17. Project Phases & Timeline
 - 18. Acceptance Criteria
 - 19. Assumptions & Exclusions
 - 20. Future Scope
-

1. Project Overview

The Play School Digital Ecosystem is a multi-surface platform designed for early-childhood education institutions. It consists of:

1. A premium, animated, responsive **Public Website**
2. A **Parent Mobile Application** (iOS + Android)
3. A **Child Interactive Learning Interface** (educational games + content)
4. A **School Administration Dashboard** (multi-branch, multi-role)
5. A **Teacher & Staff Management Panel**

The product is positioned as a **complete digital learning and engagement ecosystem for early childhood development** — not a traditional school portal. It blends Disney-style emotional warmth, modern EdTech usability, interactive gaming, and parent trust-building UX.

2. Project Vision & Goals

2.1 Vision

To deliver a warm, safe, interactive, playful, intelligent, educational, emotionally engaging, and visually immersive ecosystem for play schools — built to scale across branches, franchises, and geographies.

2.2 Strategic Goals

- Build a brand-differentiating digital identity for the play school
- Centralize parent communication, fee management, and child progress reporting
- Provide age-adaptive educational games and learning modules for children
- Enable multi-branch administration with rich analytics
- Architect the platform for AI personalization, AR learning, and online classes in future phases

2.3 Success Metrics

- 90%+ parent app adoption within 60 days of launch
 - < 2.5s LCP on website, < 3s cold start on mobile app
 - 95%+ uptime SLA
 - Daily active engagement on child learning modules
 - Reduction in manual fee/attendance/communication workload at branch level
-

3. Master Feature List

3.1 Brand & Design Direction

- Modern glassmorphism + soft UI with rounded edges
- Child-safe color palette: Sky Blue, Sunshine Yellow, Soft Orange, Mint Green, Lavender Purple, Coral Pink, Soft White

- Motion-driven engagement (Framer Motion + GSAP)
- 1–3 mascots used across onboarding, achievements, navigation, and games
- Seasonal theming (Christmas, Diwali, Halloween, Independence Day, Summer Camp, anniversary)

3.2 Public Website

- **Home:** Hero animation, mascot welcome, about, why-choose-us, methodology, virtual campus preview, activities showcase, testimonials, teachers, growth programs, admission CTA, events, safety standards, parent app showcase, games showcase, interactive FAQ, contact & map
- **About Us:** Vision/mission, founder message, philosophy, teaching approach, child psychology, emotional development, learn-through-play, interactive timeline, animated storytelling
- **Programs & Curriculum:** Toddlers, Nursery, Junior KG, Senior KG, Activity programs, Summer camps, Weekend workshops — each with outcomes, activities, skill development, interactive preview, visual timeline, parent expectations, milestones
- **Admissions:** Online inquiry, application, document uploads, fee structure, branch selection, seat availability, workflow tracking, visit scheduler, video counseling booking, lead nurturing automation
- **Interactive Campus Tour:** 360° classroom walkthrough, animated map, virtual guided tour, clickable hotspots, storytelling mode, teacher intro popups, facility highlights
- **Events & Activities:** Calendar, photo/video galleries, annual day, sports, art competitions, parent participation, festivals, RSVP
- **Parent Testimonials:** Video testimonials, ratings, parent stories, interactive sliders, progress experiences
- **Blog:** Parenting tips, child psychology, nutrition, learning methods, activity ideas, screen-time management; SEO, social sharing, AI recommendations, tagging, reading progress
- **Contact:** Smart form, WhatsApp, Google Maps, branch selector, callback requests, emergency contact, social media

3.3 Parent Mobile Application

- **Parent Dashboard:** Child overview, attendance summary, daily updates, homework, notifications, upcoming events, meal/sleep/activity/mood reports, growth tracking, fee reminders, performance snapshots
- **Daily Activity Feed:** Photo/video posts, teacher comments, learning updates, participation highlights, emotional notes, timestamps
- **Attendance:** Live status, check-in/check-out logs, pickup authorization, guardian tracking, real-time alerts, QR pickup

- **Notice Board:** Circulars, announcements, event notices, holidays, exam/activity updates, push notifications, rich media
- **Payments:** Online fees, auto reminders, installments, receipts, invoices, multiple gateways, UPI, subscription plans, financial history, late fee automation
- **Parent-Teacher Communication:** Secure messaging, voice notes, video consultation booking, feedback, scheduled meetings, activity/performance discussions
- **Child Progress Reports:** Development analytics, cognitive tracking, skill growth, activity participation, emotional indicators, interactive charts, milestones, printable reports
- **Emergency & Safety:** Emergency contact, live alerts, medical info, allergy tracking, emergency pickup, safety broadcasts

3.4 Child Learning & Gaming Ecosystem

Age-segmented, safe, reward-based, audio-assisted, parent-controlled.

- **Age 2–3:** Balloon pop, shape drag-drop, animal sound matching, color sorting, bubble interactions
- **Age 3–4:** Alphabet tracing, matching cards, counting games, musical learning, puzzle assembly
- **Age 4–5:** Word builder, interactive stories, math adventures, pattern recognition, story sequencing
- **Age 5–6:** Educational quests, mini science labs, creative drawing, music creation, world exploration

3.5 Gamification Engine

- Reward system: stars, badges, trophies, unlockable characters, daily streaks, skill badges, learning levels
- Learning journey map (adventure-style progression unlocking levels, worlds, mascots, sounds, mini-games)
- Audio interaction: voice-guided instructions, pronunciation support, musical feedback, narration, storytelling, multi-language

3.6 Interactive Learning Modules

- Storytelling (animated, read-along, choice-based, narration, moral stories, regional languages, AI voices)
- Drawing & Creativity (finger painting, coloring books, shape drawing, save artwork, stickers, rewards)
- Music & Rhythm (instruments, rhythm tapping, musical puzzles, rhymes, karaoke)
- AR Learning (future scope): AR animals, alphabets, planets, camera learning, gestures

3.7 Teacher Management

- Teacher dashboard (attendance, activity updates, report creation, homework assignment, communication, scheduling, performance analytics)
- Classroom activity uploads (image/video, tagging, parent sharing permissions, moderation)
- Lesson planning (curriculum, daily lessons, resources, scheduling, goal tracking)

3.8 Administration Dashboard

- Super Admin: multi-branch, role management, student/staff database, payments, analytics, inquiry tracking, admission pipeline, notification controls, app management
- Advanced analytics: attendance, fees, engagement, parent activity, game usage, learning progress, retention
- CMS: dynamic content, blog, homepage, banners, announcements, SEO, media library

3.9 AI & Smart Features (Future Ready)

- Personalized learning paths and recommendations
- Adaptive difficulty
- AI parent assistant (FAQs, parenting tips, child summaries, smart reminders)
- AI child analytics (behavior insights, engagement, trends)

3.10 Marketing & Growth

- SEO (local, structured data, blog, fast indexing, landing pages)
- Lead generation (funnels, retargeting, CRM, automated follow-ups, WhatsApp automation)
- Referral system (rewards, tracking, campaigns)

3.11 Engagement Systems

- Multi-channel notifications (push, email, SMS, WhatsApp, in-app)
- Seasonal experience theming

3.12 Media & Content System

- Secure image hosting, video compression, AI tagging, gallery management, story highlights, event albums
-

4. Technology Stack

4.1 Frontend (Website)

Layer	Technology
Framework	Next.js 16 (App Router, RSC)
Language	TypeScript 5
UI Library	React 19
Styling	Tailwind CSS 4, CSS Modules
Component Library	shadcn/ui
Animation	Framer Motion, GSAP, Lottie
3D / Interactive	Three.js / React Three Fiber
State	Zustand
Data Fetching	TanStack Query, Axios
Forms	React Hook Form + Zod
Icons	Lucide, custom SVG mascots

4.2 Mobile App (Parent + Child)

Layer	Technology
Framework	React Native (Expo SDK 53)
Navigation	Expo Router / React Navigation
State	Zustand + TanStack Query
Animation	Reanimated 3, Moti, Lottie, Skia
Audio	expo-av
Camera/QR	expo-camera, expo-barcode-scanner
Push	Expo Notifications + FCM/APNs
Storage	MMKV, SecureStore

4.3 Backend

Layer	Technology
Runtime	Node.js 22 LTS
API Layer	Next.js Route Handlers (App Router)
ORM	Prisma 7
Database	PostgreSQL (NeonDB)
Cache / Queue	Redis (Upstash) + BullMQ
Realtime	WebSocket (Pusher / Ably)

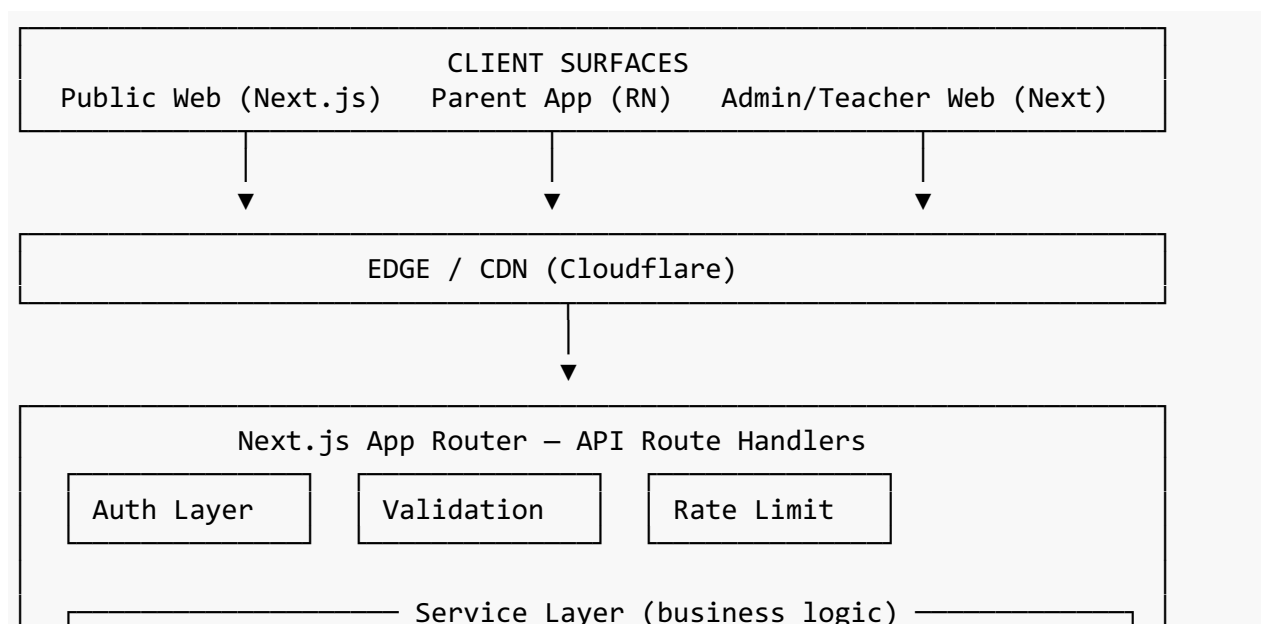
Layer	Technology
Auth	NextAuth.js (Auth.js) + JWT
Validation	Zod
File Storage	Cloudflare R2 / AWS S3
Media Processing	Sharp, FFmpeg, Mux (video)
Email	Resend
SMS	Twilio / MSG91
WhatsApp	WhatsApp Cloud API
Payments	Razorpay, Stripe

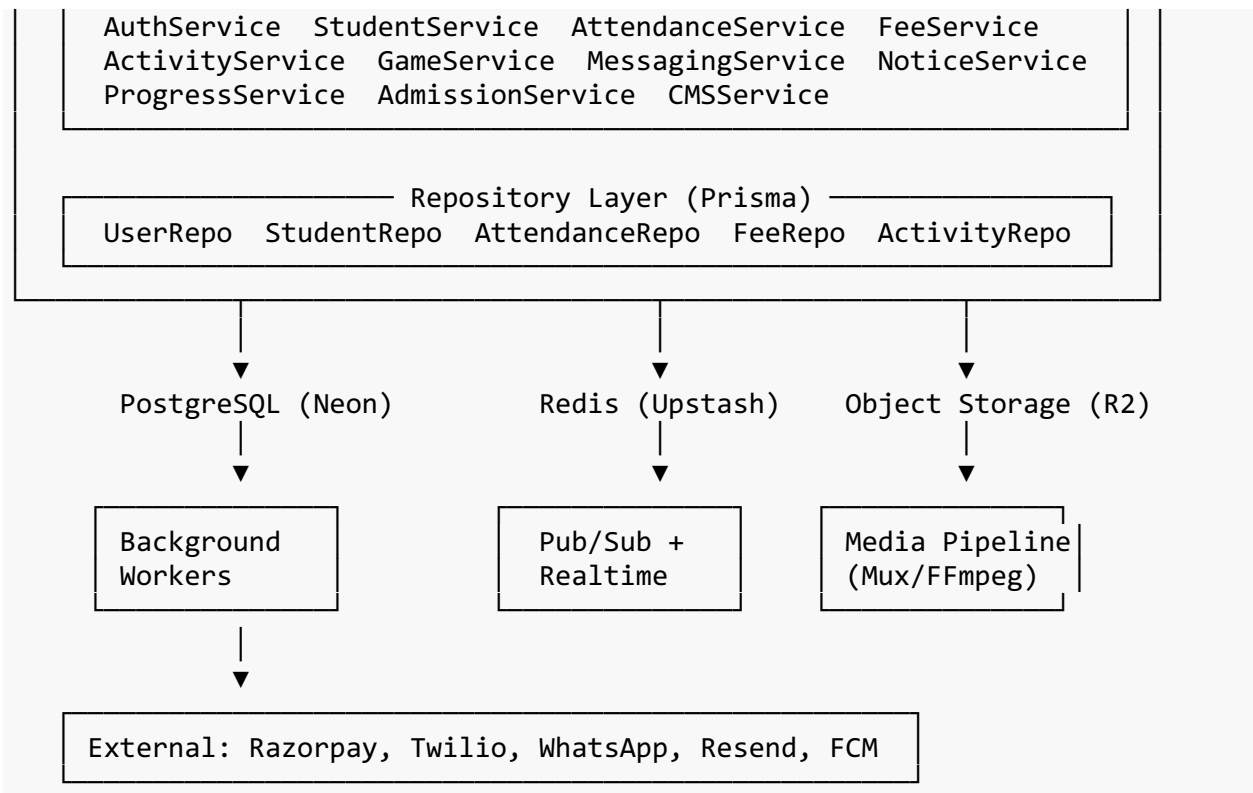
4.4 DevOps & Infra

Layer	Technology
Hosting (web)	Vercel
Hosting (mobile build)	EAS Build
CDN	Cloudflare
Monitoring	Sentry, Vercel Analytics
Logs	Logflare / Better Stack
CI/CD	GitHub Actions
Secrets	Doppler / Vercel env

5. Software Architecture

5.1 High-Level Architecture





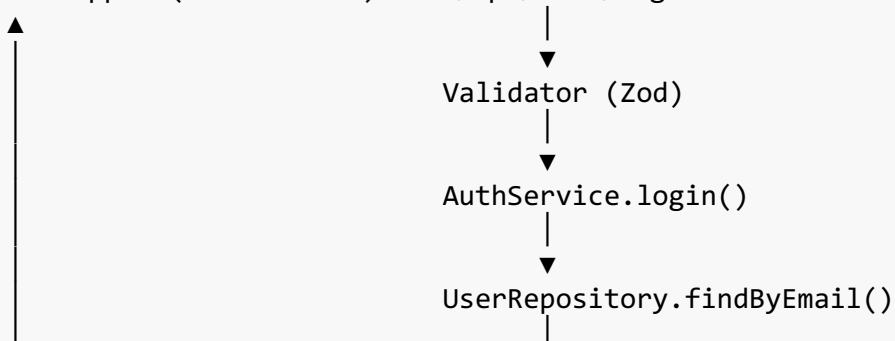
5.2 Architectural Principles

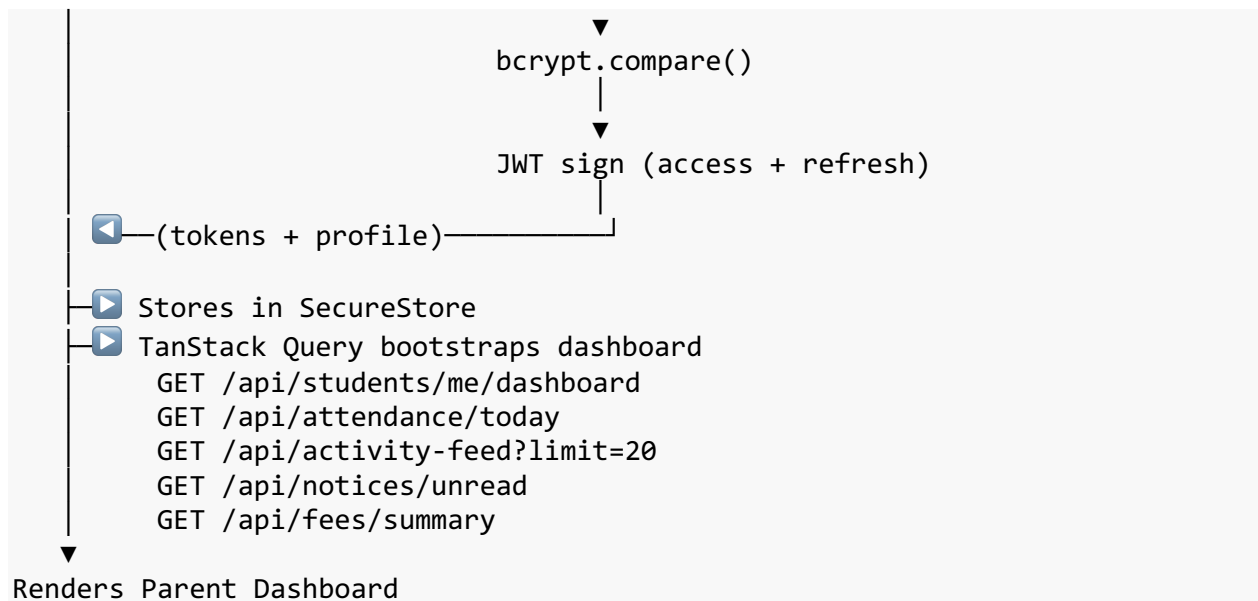
- **Layered architecture:** Route → Validator → Service → Repository → DB
- **Single source of truth** for shared types in src/shared/types
- **Stateless API** with JWT; session refresh through HTTP-only cookies for web
- **Event-driven** background jobs (notifications, media processing, reports) via BullMQ
- **Multi-tenant ready:** Branch-scoped data with row-level filtering
- **Role-based access control (RBAC)** enforced in middleware and service layer

6. Data Flow

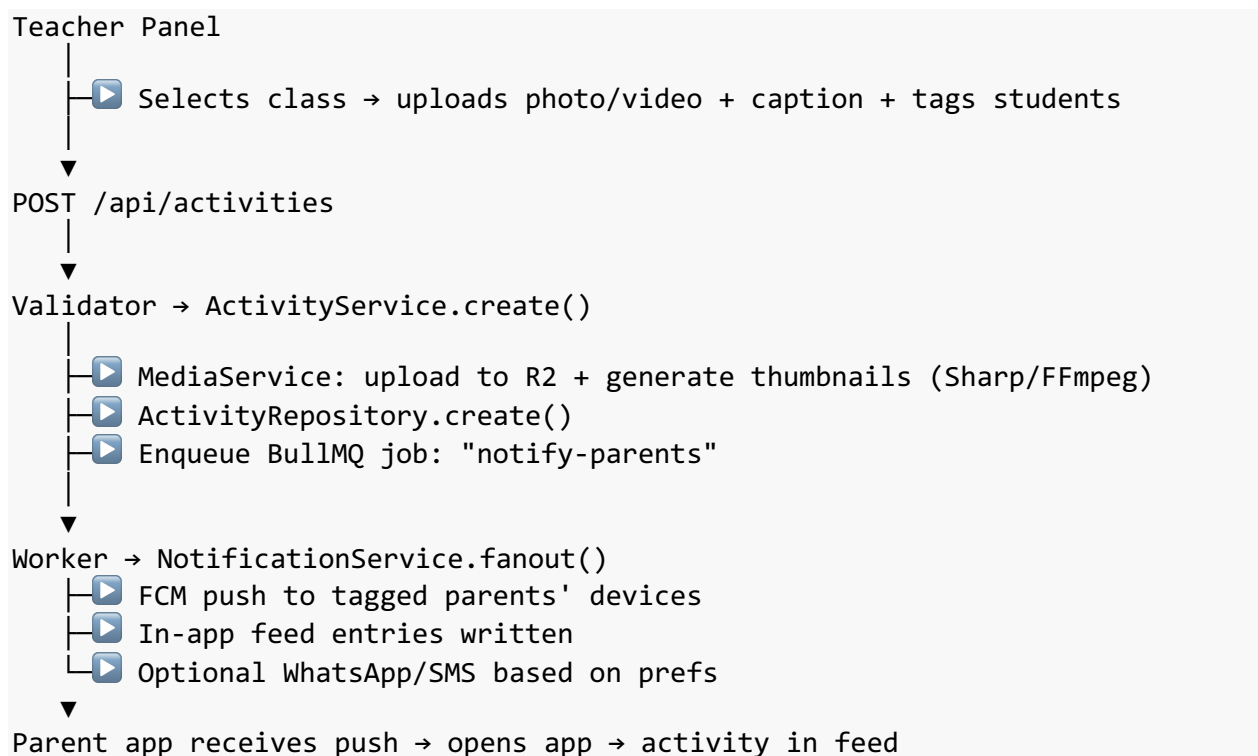
6.1 Parent Login → Dashboard Flow

Parent App —(credentials)—▶ /api/auth/login

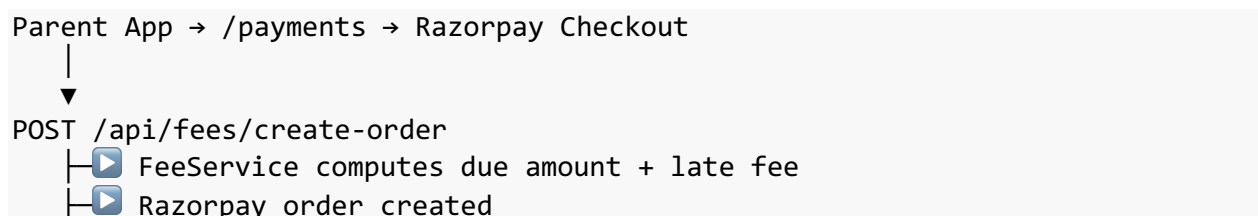


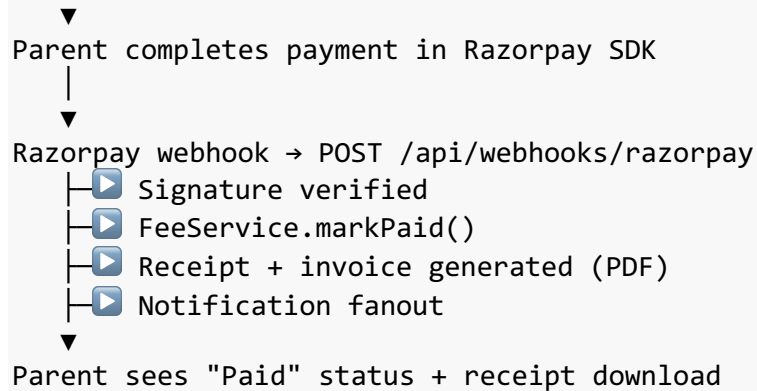


6.2 Teacher Uploads Daily Activity → Parent Sees Update

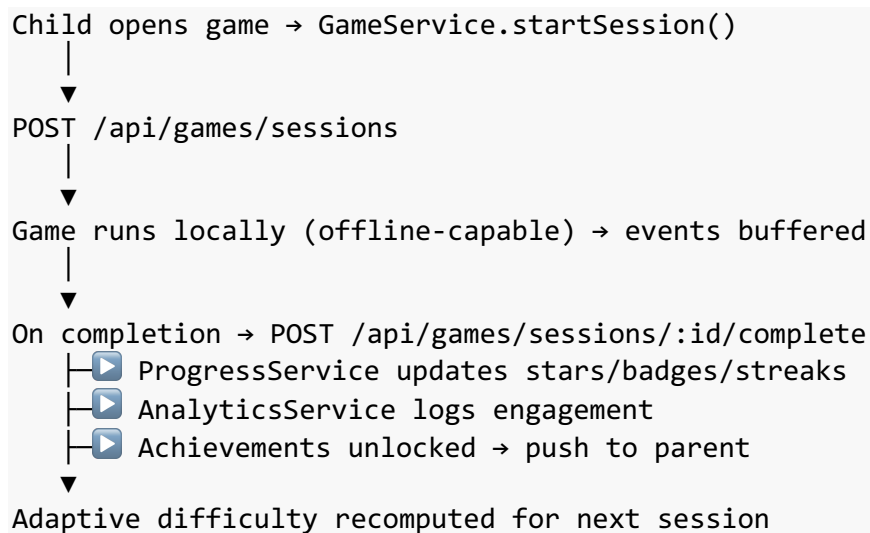


6.3 Fee Payment Flow





6.4 Child Game Session Flow



7. API Reference

All endpoints are JSON, JWT-authenticated (except /auth/*, /public/*, /webhooks/*), rate-limited, validated via Zod, and follow /api/v1/... versioning.

7.1 Authentication

Method	Endpoint	Description
POST	/api/auth/register	Parent self-register (invite code)
POST	/api/auth/login	Email + password login
POST	/api/auth/refresh	Refresh access token
POST	/api/auth/logout	Revoke refresh token
POST	/api/auth/forgot-password	Initiate reset email
POST	/api/auth/reset-password	Reset with token
POST	/api/auth/verify-otp	Verify phone OTP

Method	Endpoint	Description
GET	/api/auth/me	Current user profile

7.2 Users & Roles

Method	Endpoint	Description
GET	/api/users	List users (admin)
GET	/api/users/:id	User detail
POST	/api/users	Create user (admin)
PATCH	/api/users/:id	Update user
DELETE	/api/users/:id	Soft-delete user
GET	/api/roles	Role list
POST	/api/roles/assign	Assign role to user

7.3 Branches

Method	Endpoint	Description
GET	/api/branches	List branches
POST	/api/branches	Create branch
PATCH	/api/branches/:id	Update branch

7.4 Students

Method	Endpoint	Description
GET	/api/students	List students (branch-scoped)
GET	/api/students/:id	Student detail
POST	/api/students	Enroll student
PATCH	/api/students/:id	Update student
GET	/api/students/:id/dashboard	Parent dashboard payload
GET	/api/students/:id/progress	Progress report
GET	/api/students/:id/milestones	Milestone log

7.5 Admissions

Method	Endpoint	Description
POST	/api/admissions/inquiry	Public inquiry submission
POST	/api/admissions/applications	Create application
GET	/api/admissions/applications	List (admin)
PATCH	/api/admissions/applications/:id	Update stage
POST	/api/admissions/applications/:id/documents	Upload docs
GET	/api/admissions/seats	Seat availability

Method	Endpoint	Description
POST	/api/admissions/visit-bookings	Schedule school visit
POST	/api/admissions/counseling	Book video counseling

7.6 Attendance

Method	Endpoint	Description
POST	/api/attendance/check-in	Teacher/QR check-in
POST	/api/attendance/check-out	Check-out + pickup person
GET	/api/attendance/today	Today's status
GET	/api/attendance/student/:id	Per-student history
GET	/api/attendance/class/:id	Per-class roster
POST	/api/attendance/pickup-authorization	Authorize pickup

7.7 Activity Feed

Method	Endpoint	Description
GET	/api/activities	Feed (filtered)
POST	/api/activities	Teacher creates post
PATCH	/api/activities/:id	Edit
DELETE	/api/activities/:id	Remove
POST	/api/activities/:id/comments	Comment
POST	/api/activities/:id/reactions	React

7.8 Notices

Method	Endpoint	Description
GET	/api/notices	List notices
POST	/api/notices	Create notice (admin/teacher)
GET	/api/notices/:id	Detail
PATCH	/api/notices/:id	Update
POST	/api/notices/:id/read	Mark read

7.9 Messaging

Method	Endpoint	Description
GET	/api/conversations	Threads
POST	/api/conversations	Start thread
GET	/api/conversations/:id/messages	Messages
POST	/api/conversations/:id/messages	Send (text/voice/file)
POST	/api/conversations/:id/typing	Typing indicator

Method	Endpoint	Description
POST	/api/meetings	Book video consult

7.10 Fees & Payments

Method	Endpoint	Description
GET	/api/fees/summary	Outstanding + upcoming
GET	/api/fees/invoices	Invoice list
GET	/api/fees/invoices/:id	Detail + PDF
POST	/api/fees/create-order	Create payment order
POST	/api/fees/installments	Set plan
GET	/api/fees/receipts/:id	Receipt PDF
POST	/api/webhooks/razorpay	Razorpay webhook
POST	/api/webhooks/stripe	Stripe webhook

7.11 Events & Calendar

Method	Endpoint	Description
GET	/api/events	List events
POST	/api/events	Create event
POST	/api/events/:id/rsvp	RSVP
GET	/api/events/:id/gallery	Photo/video gallery

7.12 Curriculum & Programs

Method	Endpoint	Description
GET	/api/programs	Program list (toddlers/nursery/JKG/SKG)
GET	/api/programs/:slug	Program detail
GET	/api/curriculum/:programId	Curriculum timeline
GET	/api/lessons	Lesson list (teacher)
POST	/api/lessons	Create lesson plan
PATCH	/api/lessons/:id	Update lesson

7.13 Child Learning & Games

Method	Endpoint	Description
GET	/api/games	Game catalog (by age)
GET	/api/games/:slug	Game detail + manifest
POST	/api/games/sessions	Start session
POST	/api/games/sessions/:id/complete	Submit results
GET	/api/progress/:studentId	Learning progress

Method	Endpoint	Description
GET	/api/achievements/:studentId	Achievements + badges
GET	/api/journey/:studentId	Adventure-map state
GET	/api/stories	Storytelling content
GET	/api/stories/:id	Story detail + audio

7.14 Emergency & Safety

Method	Endpoint	Description
POST	/api/emergency/contact	Trigger contact request
GET	/api/emergency/broadcasts	Safety broadcasts
GET	/api/medical/:studentId	Medical/allergy info
PATCH	/api/medical/:studentId	Update

7.15 CMS (Admin)

Method	Endpoint	Description
GET	/api/cms/pages	List pages
PATCH	/api/cms/pages/:slug	Edit dynamic page
GET	/api/cms/blog	Blog posts
POST	/api/cms/blog	Create post
GET	/api/cms/banners	Banners
GET	/api/cms/media	Media library

7.16 Analytics

Method	Endpoint	Description
GET	/api/analytics/attendance	Attendance trends
GET	/api/analytics/fees	Fee collection
GET	/api/analytics/engagement	App engagement
GET	/api/analytics/games	Game usage
GET	/api/analytics/learning	Learning progress
GET	/api/analytics/retention	Retention cohorts

7.17 Notifications

Method	Endpoint	Description
GET	/api/notifications	List for current user
POST	/api/notifications/read	Mark read
POST	/api/notifications/devices	Register device token
POST	/api/notifications/preferences	Update channels

7.18 Public/Marketing

Method	Endpoint	Description
POST	/api/public/contact	Contact form
POST	/api/public/callback	Callback request
POST	/api/public/newsletter	Newsletter signup
GET	/api/public/testimonials	Testimonials
POST	/api/public/referrals	Referral capture

8. Route Map

8.1 Public Website Routes

Route	Page
/	Home (hero, mascot, showcases)
/about	About Us
/programs	Programs index
/programs/[slug]	Program detail (toddlers, nursery, JKG, SKG, summer-camp)
/admissions	Admissions landing
/admissions/apply	Application form
/admissions/visit	Book visit
/admissions/seats	Seat availability
/campus-tour	Interactive 360° tour
/events	Events list
/events/[slug]	Event detail
/gallery	Photo/video gallery
/testimonials	Testimonials
/blog	Blog index
/blog/[slug]	Blog post
/contact	Contact + map
/careers	Careers

8.2 Auth Routes

Route	Page
/login	Login
/register	Register (invite code)

Route	Page
/forgot-password	Reset
/verify-otp	OTP verification

8.3 Parent Portal Routes (Web Mirror of Mobile App)

Route	Page
/parent	Dashboard
/parent/feed	Activity feed
/parent/attendance	Attendance
/parent/notices	Notice board
/parent/messages	Messages
/parent/payments	Fees
/parent/payments/invoices	Invoices
/parent/reports	Progress reports
/parent/events	Events
/parent/emergency	Emergency contacts
/parent/profile	Profile
/parent/settings	Settings

8.4 Child Learning Zone Routes

Route	Page
/kids	Kids home (mascot picker)
/kids/journey	Adventure map
/kids/games	Games catalog
/kids/games/[slug]	Game player
/kids/stories	Storytelling
/kids/stories/[id]	Story player
/kids/draw	Drawing & creativity
/kids/music	Music & rhythm
/kids/rewards	Badges & trophies

8.5 Teacher Panel Routes

Route	Page
/teacher	Dashboard
/teacher/classes	Class list
/teacher/classes/[id]	Class detail

Route	Page
/teacher/attendance	Attendance marker
/teacher/activities	Activity uploads
/teacher/lessons	Lesson planner
/teacher/messages	Messages
/teacher/reports	Student reports

8.6 Admin Dashboard Routes

Route	Page
/admin	Overview
/admin/branches	Branches
/admin/students	Students
/admin/staff	Staff
/admin/admissions	Admissions pipeline
/admin/fees	Fee management
/admin/notices	Notice management
/admin/events	Events
/admin/cms	CMS
/admin/cms/blog	Blog editor
/admin/cms/media	Media library
/admin/analytics	Analytics
/admin/roles	Roles & permissions
/admin/settings	Settings
/admin/audit	Audit logs

8.7 Mobile App (Parent) Screens

Screen	Description
Splash	Branded splash + mascot
Onboarding	First-time intro slides
Login	Email/phone login
Dashboard	Child overview
Feed	Daily activity feed
Attendance	Attendance & pickup
Notices	Notice board
Messages	Chat
Payments	Fees

Screen	Description
Reports	Progress
Profile	Parent profile
Settings	Settings + notifications
KidsZone	Child learning entry

9. Folder Structure

```

playschool/
├── src/
│   └── app/                                     # Next.js App Router (Frontend)
│       ├── (auth)/
│       │   ├── login/
│       │   │   └── page.tsx
│       │   ├── register/
│       │   │   └── page.tsx
│       │   ├── forgot-password/
│       │   │   └── page.tsx
│       │   ├── verify-otp/
│       │   │   └── page.tsx
│       │   └── layout.tsx
│       ├── (marketing)/
│       │   ├── about/page.tsx
│       │   ├── programs/page.tsx
│       │   ├── programs/[slug]/page.tsx
│       │   ├── admissions/page.tsx
│       │   ├── admissions/apply/page.tsx
│       │   ├── admissions/visit/page.tsx
│       │   ├── campus-tour/page.tsx
│       │   ├── events/page.tsx
│       │   ├── events/[slug]/page.tsx
│       │   ├── gallery/page.tsx
│       │   ├── testimonials/page.tsx
│       │   ├── blog/page.tsx
│       │   ├── blog/[slug]/page.tsx
│       │   ├── contact/page.tsx
│       │   ├── careers/page.tsx
│       │   └── layout.tsx
│       └── (parent)/
│           ├── parent/page.tsx
│           ├── parent/feed/page.tsx
│           ├── parent/attendance/page.tsx
│           ├── parent/notices/page.tsx
│           └── parent/messages/page.tsx

```

- parent/payments/page.tsx
- parent/payments/invoices/page.tsx
- parent/reports/page.tsx
- parent/events/page.tsx
- parent/emergency/page.tsx
- parent/profile/page.tsx
- parent/settings/page.tsx
- layout.tsx

— (kids)/

- kids/page.tsx
- kids/journey/page.tsx
- kids/games/page.tsx
- kids/games/[slug]/page.tsx
- kids/stories/page.tsx
- kids/stories/[id]/page.tsx
- kids/draw/page.tsx
- kids/music/page.tsx
- kids/rewards/page.tsx
- layout.tsx

— (teacher)/

- teacher/page.tsx
- teacher/classes/page.tsx
- teacher/classes/[id]/page.tsx
- teacher/attendance/page.tsx
- teacher/activities/page.tsx
- teacher/lessons/page.tsx
- teacher/messages/page.tsx
- teacher/reports/page.tsx
- layout.tsx

— (admin)/

- admin/page.tsx
- admin/branches/page.tsx
- admin/students/page.tsx
- admin/staff/page.tsx
- admin/admissions/page.tsx
- admin/fees/page.tsx
- admin/notices/page.tsx
- admin/events/page.tsx
- admin/cms/page.tsx
- admin/cms/blog/page.tsx
- admin/cms/media/page.tsx
- admin/analytics/page.tsx
- admin/roles/page.tsx
- admin/settings/page.tsx
- admin/audit/page.tsx
- layout.tsx

```
├── layout.tsx
├── page.tsx                                # Public home
├── globals.css

├── backend/                               # Backend Layer (API Logic)
│   ├── api/                              # API Routes (Next.js handlers)
│   │   ├── auth/
│   │   │   ├── login/route.ts
│   │   │   ├── register/route.ts
│   │   │   ├── refresh/route.ts
│   │   │   ├── logout/route.ts
│   │   │   ├── forgot-password/route.ts
│   │   │   ├── reset-password/route.ts
│   │   │   ├── verify-otp/route.ts
│   │   │   └── me/route.ts
│   │   ├── users/route.ts
│   │   ├── users/[id]/route.ts
│   │   ├── roles/route.ts
│   │   ├── branches/route.ts
│   │   ├── students/route.ts
│   │   ├── students/[id]/route.ts
│   │   ├── students/[id]/dashboard/route.ts
│   │   ├── students/[id]/progress/route.ts
│   │   ├── admissions/
│   │   │   ├── inquiry/route.ts
│   │   │   ├── applications/route.ts
│   │   │   ├── applications/[id]/route.ts
│   │   │   ├── seats/route.ts
│   │   │   ├── visit-bookings/route.ts
│   │   │   └── counseling/route.ts
│   │   ├── attendance/
│   │   │   ├── check-in/route.ts
│   │   │   ├── check-out/route.ts
│   │   │   ├── today/route.ts
│   │   │   └── pickup-authorization/route.ts
│   │   ├── activities/route.ts
│   │   ├── activities/[id]/route.ts
│   │   ├── notices/route.ts
│   │   ├── conversations/route.ts
│   │   ├── conversations/[id]/messages/route.ts
│   │   ├── meetings/route.ts
│   │   ├── fees/
│   │   │   ├── summary/route.ts
│   │   │   ├── invoices/route.ts
│   │   │   ├── create-order/route.ts
│   │   │   ├── installments/route.ts
│   │   │   └── receipts/[id]/route.ts
│   │   ├── events/route.ts
│   │   ├── events/[id]/rsvp/route.ts
│   │   └── programs/route.ts
```

```
— curriculum/[programId]/route.ts
— lessons/route.ts
— games/route.ts
— games/sessions/route.ts
— games/sessions/[id]/complete/route.ts
— stories/route.ts
— progress/[studentId]/route.ts
— achievements/[studentId]/route.ts
— journey/[studentId]/route.ts
— emergency/contact/route.ts
— medical/[studentId]/route.ts
— cms/pages/route.ts
— cms/blog/route.ts
— cms/banners/route.ts
— cms/media/route.ts
— analytics/
  — attendance/route.ts
  — fees/route.ts
  — engagement/route.ts
  — games/route.ts
  — learning/route.ts
  — retention/route.ts
— notifications/route.ts
— notifications/devices/route.ts
— webhooks/razorpay/route.ts
— webhooks/stripe/route.ts
— public/
  — contact/route.ts
  — callback/route.ts
  — newsletter/route.ts
  — testimonials/route.ts
  — referrals/route.ts
— middleware.ts

— services/                                # Business Logic Layer
  — auth.service.ts
  — user.service.ts
  — branch.service.ts
  — student.service.ts
  — admission.service.ts
  — attendance.service.ts
  — activity.service.ts
  — notice.service.ts
  — messaging.service.ts
  — fee.service.ts
  — payment.service.ts
  — event.service.ts
  — curriculum.service.ts
  — lesson.service.ts
  — game.service.ts
```

- progress.service.ts
- achievement.service.ts
- story.service.ts
- emergency.service.ts
- cms.service.ts
- analytics.service.ts
- notification.service.ts
- media.service.ts
- email.service.ts

- repositories/ # Data Access Layer

- auth.repository.ts
- user.repository.ts
- branch.repository.ts
- student.repository.ts
- admission.repository.ts
- attendance.repository.ts
- activity.repository.ts
- notice.repository.ts
- message.repository.ts
- fee.repository.ts
- event.repository.ts
- lesson.repository.ts
- game.repository.ts
- progress.repository.ts
- story.repository.ts
- cms.repository.ts
- notification.repository.ts

- database/ # Database Configuration

- prisma/
 - schema.prisma
 - migrations/
- client.ts
- seed.ts

- validators/ # Request Validation (Zod)

- auth.validator.ts
- user.validator.ts
- student.validator.ts
- admission.validator.ts
- attendance.validator.ts
- activity.validator.ts
- notice.validator.ts
- message.validator.ts
- fee.validator.ts
- event.validator.ts
- lesson.validator.ts
- game.validator.ts
- cms.validator.ts

- jobs/ # Background Workers (BullMQ)
 - notification.job.ts
 - media-processing.job.ts
 - report-generation.job.ts
 - fee-reminder.job.ts
 - analytics-rollup.job.ts

- utils/ # Backend Utilities
 - jwt.util.ts
 - hash.util.ts
 - rate-limit.util.ts
 - rbac.util.ts
 - pdf.util.ts
 - webhook.util.ts
 - error-handler.util.ts

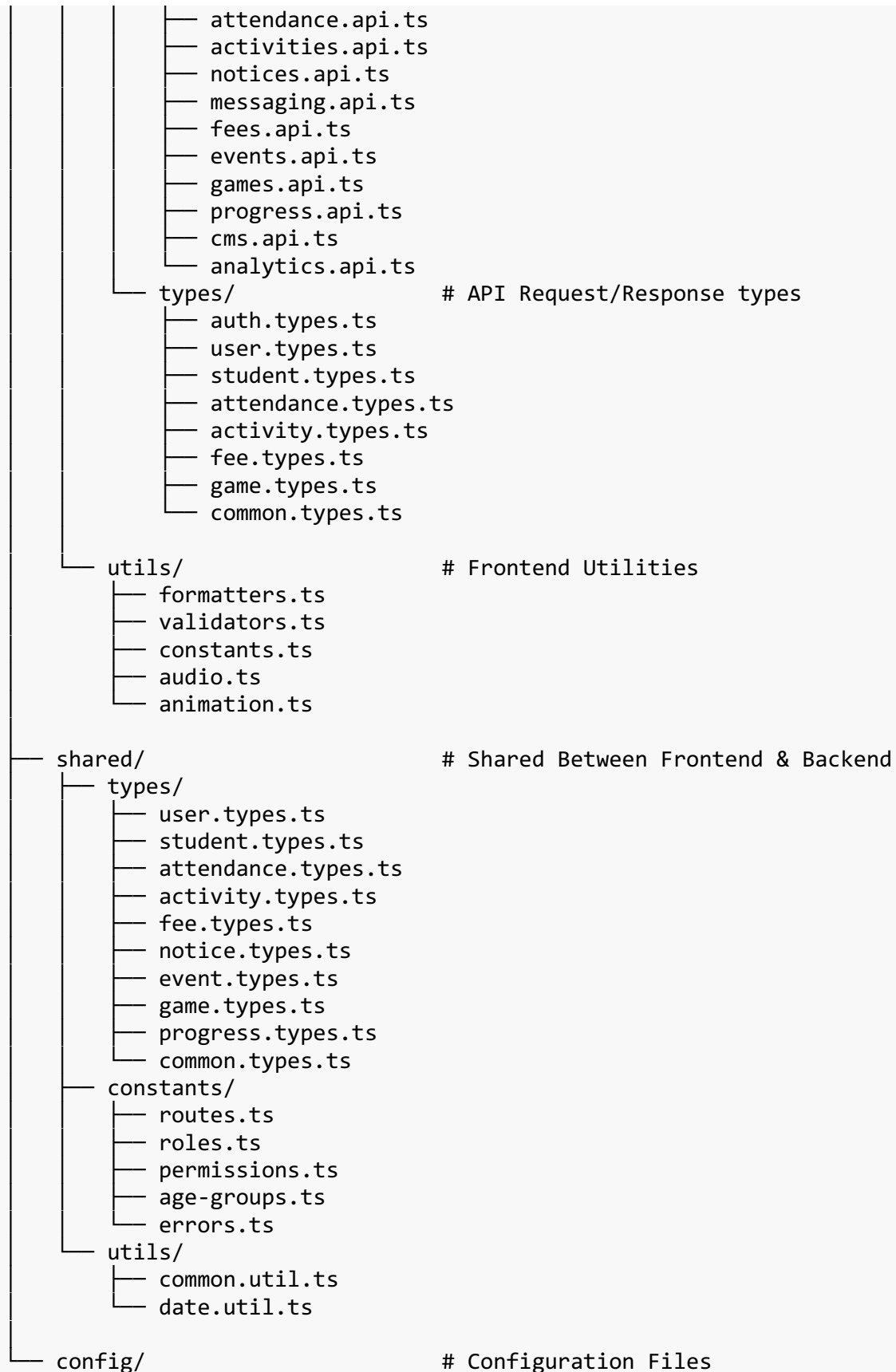
- frontend/ # Frontend-Specific Logic
 - components/ # React Components
 - ui/ # Reusable UI (shadcn)
 - Button.tsx
 - Input.tsx
 - Modal.tsx
 - Card.tsx
 - Sheet.tsx
 - Toast.tsx
 - Skeleton.tsx
 - animation/ # Motion primitives
 - FloatingMascot.tsx
 - ParallaxScene.tsx
 - ConfettiBurst.tsx
 - BalloonPop.tsx
 - PageTransition.tsx
 - mascot/ # Mascot system
 - MascotGuide.tsx
 - MascotReactor.tsx
 - mascot.config.ts
 - features/ # Feature-Specific Components
 - auth/
 - LoginForm.tsx
 - RegisterForm.tsx
 - marketing/
 - HeroScene.tsx
 - ProgramShowcase.tsx
 - TestimonialSlider.tsx
 - CampusTour360.tsx
 - parent/
 - DashboardSummary.tsx
 - ActivityFeed.tsx
 - AttendanceCard.tsx

```
├── FeeWidget.tsx
├── ProgressChart.tsx
├── kids/
│   ├── JourneyMap.tsx
│   ├── GameTile.tsx
│   ├── StoryPlayer.tsx
│   ├── DrawingCanvas.tsx
│   ├── MusicStudio.tsx
│   └── RewardLocker.tsx
├── teacher/
│   ├── ClassRoster.tsx
│   ├── ActivityUploader.tsx
│   └── LessonPlanner.tsx
├── admin/
│   ├── BranchSwitcher.tsx
│   ├── AdmissionPipeline.tsx
│   ├── AnalyticsBoard.tsx
│   └── CMSEditor.tsx
├── layout/ # Layout Components
│   ├── Header.tsx
│   ├── Sidebar.tsx
│   ├── Footer.tsx
│   └── MobileTabBar.tsx
```

```
├── hooks/ # Custom React Hooks
│   ├── useAuth.ts
│   ├── useUser.ts
│   ├── useStudent.ts
│   ├── useAttendance.ts
│   ├── useFeed.ts
│   ├── useFees.ts
│   ├── useProgress.ts
│   ├── useGames.ts
│   ├── useNotifications.ts
│   └── useMascot.ts
```

```
├── store/ # State Management (Zustand)
│   ├── authStore.ts
│   ├── userStore.ts
│   ├── appStore.ts
│   ├── kidsStore.ts
│   └── notificationStore.ts
```

```
├── api/ # Frontend API Client Layer
│   ├── client.ts # Axios instance & interceptors
│   └── endpoints/
│       ├── auth.api.ts
│       ├── users.api.ts
│       ├── students.api.ts
│       └── admissions.api.ts
```

- env.ts - api.config.ts - app.config.ts - theme.config.ts	# Environment validation (Zod)
- mobile/ - app/ - components/ - hooks/ - store/ - api/ - assets/ - app.json	# React Native (Expo) App # Expo Router routes
- public/ - images/ - icons/ - mascots/ - lottie/ - audio/	# Static Assets
- tests/ - unit/ - integration/ - e2e/	# Test Files
- docs/ - sow/ - sow.md	
.env.local .env.development .env.production - next.config.ts - tsconfig.json - tailwind.config.ts - package.json - README.md	

10. User Roles & Permissions

Role	Capabilities
Super Admin	Full system access, multi-branch, roles, analytics, CMS, billing
Branch Admin	Manage branch staff/students/fees/events, branch-scoped analytics

Role	Capabilities
Teacher	Class roster, attendance, activity uploads, messaging, lesson plans, reports
Parent	View child data, payments, messages, notices, activity feed, emergency
Student/Child	Sandboxed games, stories, drawing, music, rewards (no PII inputs)
Finance Team	Fee structure, invoices, receipts, refunds, finance analytics
Counselor	Admissions inquiries, applications, visit bookings, counseling calls
Support Team	View-only access to issues, parent profiles, app logs (read-only PII)

RBAC is enforced via middleware that decodes JWT → loads role + branch context → checks permission matrix before reaching the service layer.

11. Database Design Overview

Core entities (Prisma models):

- User, Role, Permission, UserRole
- Branch, Classroom, AgeGroup, Program, Curriculum, Lesson
- Student, Guardian, StudentGuardian, MedicalProfile
- Admission, Inquiry, VisitBooking, CounselingBooking
- Attendance, PickupAuthorization, EmergencyContact
- Activity, ActivityMedia, ActivityComment, ActivityReaction
- Notice, NoticeRead
- Conversation, Message, MessageAttachment, Meeting
- FeeStructure, Invoice, InvoiceLine, Payment, Receipt, Installment
- Event, EventRSVP, EventMedia
- Game, GameSession, GameEvent, Achievement, Badge, JourneyState
- Story, StoryProgress
- BlogPost, Banner, Page, MediaAsset
- Notification, DeviceToken, NotificationPreference
- AuditLog, Session

Indexes target high-traffic queries: branch + date ranges (attendance, fees), parent → child lookups, activity feed pagination, game leaderboards.

12. Security & Compliance

- HTTPS + HSTS everywhere; Cloudflare WAF
- JWT (access 15 min, refresh 30 days, rotating); HTTP-only cookies on web
- Bcrypt password hashing (cost 12); pepper from env
- Zod validation at every API boundary
- Rate limiting (Upstash) per IP + per user
- RBAC + branch scoping in service layer
- Audit logs for sensitive actions (admin, fee, role changes, exports)
- Encrypted media URLs (signed, short-TTL)
- COPPA/GDPR-inspired protections: parental consent, child PII minimization, no third-party trackers in kids zone
- Device management + session revocation
- Secrets via Doppler/Vercel env; no secrets in repo
- Penetration testing before go-live

13. Performance & Scalability

- LCP target < 2.5s, TTI < 3.5s on 4G
- ISR for marketing pages, RSC for data-heavy parent pages
- Image optimization via next/image + Cloudflare Polish
- Video via Mux (adaptive bitrate)
- Redis caching for feeds, dashboards, summaries
- Database indexing + connection pooling (PgBouncer / Neon pooler)
- BullMQ workers for fanout, media, reports
- Multi-region edge deployment (Vercel) + read replicas (future)
- Mobile: offline cache for games, lessons, last feed snapshot

14. Accessibility & Localization

- WCAG 2.1 AA target
- Large touch targets (min 44×44) on kids surfaces
- Voice assistance and narration in kids zone
- Dyslexia-friendly font option
- High-contrast theme
- Multi-language: English, Hindi, regional (configurable per branch); i18n via next-intl

15. Third-Party Integrations

Category	Provider
Payments	Razorpay (primary), Stripe (secondary)
Email	Resend
SMS	MSG91 / Twilio
WhatsApp	WhatsApp Cloud API
Push	FCM + APNs (via Expo)
Video Consult	Daily.co / Jitsi
Maps	Google Maps
Storage	Cloudflare R2
Video Hosting	Mux
Analytics	PostHog + Vercel Analytics
Error Tracking	Sentry
CRM (optional)	HubSpot / Zoho

16. Deliverables

1. Production-ready public website (Next.js)
 2. Parent mobile application (iOS + Android, Expo EAS builds + store submissions)
 3. Child learning zone (web + mobile)
 4. Teacher panel (web)
 5. Admin dashboard (web)
 6. API backend (Next.js route handlers + Prisma + Postgres)
 7. CMS (admin-managed)
 8. CI/CD pipeline (GitHub Actions → Vercel + EAS)
 9. Source code repository with branching strategy
 10. Documentation: README, API docs (OpenAPI), runbooks
 11. Training session for school admins and teachers
 12. Post-launch support window (30/60/90 days configurable)
-

17. Project Phases & Timeline

Phase	Duration	Highlights
Phase 0 — Discovery & Design	2 weeks	Brand workshop, design system, mascot creation, wireframes

Phase	Duration	Highlights
Phase 1 — Core Backend & Auth	3 weeks	Schema, auth, RBAC, branches, students
Phase 2 — Public Website	3 weeks	Marketing site, admissions, blog, campus tour
Phase 3 — Parent Portal (Web + Mobile)	4 weeks	Dashboard, feed, attendance, notices, messaging
Phase 4 — Fees & Payments	2 weeks	Razorpay, invoicing, receipts, reminders
Phase 5 — Teacher & Admin Panels	3 weeks	Activity uploads, lessons, admissions pipeline, CMS
Phase 6 — Child Learning Zone	4 weeks	Games (4 age tiers), stories, drawing, music, rewards, journey map
Phase 7 — Analytics, Notifications, Hardening	2 weeks	Analytics, push/email/SMS/WhatsApp, security review
Phase 8 — QA, UAT, Launch	2 weeks	E2E tests, store submissions, training, go-live

Total: ~25 weeks (≈6 months) with parallelizable streams.

18. Acceptance Criteria

- All features in §3 demonstrable in staging with seeded data
 - All API endpoints return correct responses with Zod-validated payloads
 - Auth + RBAC enforced; no privilege escalation possible in pen test
 - Mobile builds pass App Store + Play Store review
 - Performance budgets met (LCP, TTI, bundle size)
 - Accessibility audit (axe + manual) shows no critical issues
 - Payment flow tested end-to-end including webhook reconciliation
 - Training sessions completed for admin + teacher cohorts
-

19. Assumptions & Exclusions

Assumptions

- School provides brand assets, mascot direction, and content (photos, videos, copy, testimonials)
- Razorpay/Stripe accounts and WhatsApp Cloud API approval provided by client
- Domain, SSL, and store developer accounts (Apple + Google) procured by client
- Branch list and role hierarchy defined upfront

Exclusions (Out of Scope for v1)

- AR learning modules (planned in future phase)
 - Live online classes (planned in future phase)
 - IoT/biometric attendance hardware integration
 - Facial recognition pickup
 - Smart classroom device integrations
 - Merchandise e-commerce
 - Custom-shot video production for campus tour
-

20. Future Scope

- AI Personalization Engine (adaptive paths, recommendations)
 - AI Parent Assistant (chat + summaries)
 - AI Child Analytics (behavior, engagement, trend insights)
 - Live online classes with breakout rooms
 - AR learning (animals, alphabets, planets)
 - Smart classroom device sync
 - Subscription-based premium learning content
 - Marketplace for third-party educational content
 - White-label franchise edition
-

End of Scope of Work — Play School Digital Ecosystem v1.0